

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Command of Evidence

01

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Q1**C**

- C. Evidence emerges that the amount of iron in some rocky planets is considerably higher than the amount in their host star.

Choice C is the best answer because it presents a finding that, if true, would weaken the astronomers' claim about the makeup of host stars and their planets. The text explains that because stars and planets begin forming from the same gas and dust, astronomers believe planets should be composed of the same materials as their host stars, but in equal or smaller quantities. The finding that the amount of iron in some rocky planets is much higher than the amount in their host star would weaken the astronomers' claim because it would show that some planets contain the same material as their host star, but in higher quantities.

Choice A is incorrect because a finding only about the makeup of stars, whether they've cooled or not, would provide no information about the makeup of planets. Thus, it wouldn't have any bearing on the claim that planets and their host stars are composed of the same materials in differing quantities. Choice B is incorrect because a finding about two host stars having similar proportions of certain materials wouldn't provide any information about the makeup of planets. Thus, it wouldn't be relevant to the claim that planets and their host stars are composed of the same materials in differing quantities. Choice D is incorrect because the text indicates that the astronomers' claim is based on a fact—that stars and planets begin forming from the same gas and dust in space—which would remain true regardless of the effectiveness of a method for analysis of compositions. The text does cite analysis of rocky planets in our solar system and the Sun, but only as a single piece of evidence that is consistent with the claim and not as the source of the claim; the finding that the method used for that analysis is less effective in other scenarios wouldn't weaken a claim that's based on knowledge of how stars and planets initially form.

Q2

B

- B. If baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas it will increase 9.0% for surface water and 7.9% for groundwater if baseline precipitation is evenly distributed.

Choice B is the best answer because it describes data from the table that support Persad and her colleagues' conclusion. The text explains that, according to some climate models, precipitation in the western United States will become concentrated into fewer, more intense rain and snow events. According to the text, Persad and her colleagues concluded that more irrigation will consequently be needed but that the change in irrigation output will be highly sensitive to, or greatly affected by, the baseline concentration of precipitation in an area. This conclusion is supported by data from the researchers' simulations of changes in annual irrigation output in two different scenarios—one in which an area's annual precipitation is already somewhat concentrated and one in which its annual precipitation is evenly distributed. The table shows that if baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas if baseline precipitation is evenly distributed, water use for irrigation will increase much more—9.0% for surface water and 7.9% for groundwater. This difference illustrates the researchers' conclusion that the amount of additional water needed for irrigation will vary greatly depending on how concentrated or spread out the annual precipitation in an area already is.

Choice A is incorrect because it compares changes in the amount of water being used for irrigation to changes in the amount of water entering aquifers. Persad and her colleagues' conclusion doesn't focus on changes to the amount of water entering aquifers; rather, the researchers' conclusion focuses on changes to irrigation output relative to how concentrated or spread out the annual precipitation in an area is. Choice C is incorrect because it supports only part of Persad and her colleagues' conclusion. According to the text, the researchers concluded that the concentration of precipitation into fewer events will trigger more irrigation but that this change in irrigation output will be highly sensitive to an area's baseline concentration of annual precipitation. The data in this choice support the idea that more irrigation will be needed, but to support the rest of the researchers' conclusion, additional data from the table are required to show that the increases in water use for irrigation will vary depending on how concentrated or spread out the annual precipitation in an area already is. Choice D is incorrect because data in the table indicate no declines in water use for irrigation, showing only increases in the form of positive values.

Q3

C

- C. The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.
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Choice C is the best answer because it describes data from the table that support the researcher's hypothesis. According to the text, the researcher hypothesized that Arctic ground squirrels would exhibit longer torpor bouts and shorter arousal episodes than Alaska marmots do—or, put the other way, that the marmots would show shorter torpor bouts and longer arousal episodes than the ground squirrels do. The table shows data about torpor bouts and arousal episodes for the two species from 2008 to 2011. According to the table, the average duration of torpor bouts was 13.81 days for Alaska marmots, shorter than the average of 16.77 days for Arctic ground squirrels, and the average duration of arousal episodes was 21.2 hours for Alaska marmots, longer than the average of 14.2 hours for Arctic ground squirrels. Thus, the table supports the researcher's hypothesis by showing that Alaska marmots had shorter bouts of torpor and longer arousal episodes than Arctic ground squirrels did.

Choice A is incorrect because it inaccurately describes data from the table and doesn't support the researcher's hypothesis. The table shows that the average duration of arousal episodes was less than a day for both Alaska marmots (21.2 hours) and Arctic ground squirrels (14.2 hours). Additionally, information about arousal episodes for Alaska marmots and Arctic ground squirrels isn't sufficient to support a hypothesis involving comparisons of both arousal episodes and torpor bouts for those animals. Choice B is incorrect because it doesn't support the researcher's hypothesis, which involves comparisons of arousal episodes as well as torpor bouts for Alaska marmots and Arctic ground squirrels. Noting that both animals had torpor bouts lasting several days, on average, doesn't address arousal episodes at all, nor does it reveal how the animals' torpor bouts compared. Choice D is incorrect because it doesn't support the researcher's hypothesis. Although the table does show that Alaska marmots had more torpor bouts (12) than arousal episodes (11) and that their arousal episodes were much shorter than their torpor bouts (21.2 hours and 13.81 days, respectively), comparing data across only Alaska marmot behaviors isn't sufficient to support a hypothesis about torpor and arousal behaviors of both Alaska marmots and Arctic ground squirrels.

Q4

D

- D. Analysis of water's molecular structure under high pressure reveals that hydrogen bonds are more stable when TMAO is present than when it is not.
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Choice D is the best answer because it presents a finding that, if true, would support the researchers' hypothesis that TMAO reduces water's compressibility. The text explains that at great depths in the ocean, extreme pressure compresses the molecular structure of water by destabilizing the hydrogen bonds between adjacent molecules, thereby allowing water to penetrate proteins and harm the associated organisms. However, deep-sea organisms called piezophiles have adapted to live at these depths and previous studies show a positive correlation between the depth at which a piezophile species lives and the species' level of the compound TMAO. Because this hypothesis links TMAO levels with reduced compressibility of water's tetrahedral molecular structure, a finding that TMAO helps maintain the hydrogen bonds between water molecules under high pressure would strongly support that hypothesis.

Choice A is incorrect. Although the researchers' hypothesis suggests a relationship between TMAO and water molecules' tetrahedral molecular structure, that relationship involves TMAO helping maintain water's tetrahedral molecular structure under high pressure; as presented in the text, the hypothesis doesn't contend that water molecules are impervious to, or incapable of being penetrated by, TMAO. Choice B is incorrect because the text discusses how the molecular structure of water, not TMAO, is compressed under extreme pressure and never addresses how TMAO might be affected by such pressure. Choice C is incorrect because the researchers' hypothesis holds that water under extreme pressure is more resistant, not less, to being compressed when TMAO concentrations are higher. Moreover, the positive correlation mentioned in the text is between TMAO concentrations and the depths at which piezophiles live, not between concentrations of TMAO and the rate at which water's molecular structure compresses as pressure increases.

Q5

A

- A. Tanzania has approximately 61 million Swahili speakers, which is much more than the estimated total number of people worldwide for whom Swahili is their first language.
-

Choice A is the best answer because it most effectively uses data from the table to support the underlined claim. The text indicates that Swahili is the first language of up to 15 million people worldwide. The text goes on to claim, in the underlined portion, that even in countries where nearly everyone speaks Swahili, many of the language's speakers don't have Swahili as their first language. The table indicates that 61 million people in Tanzania, which amounts to 100 percent of the population, speak Swahili. If 61 million people in Tanzania speak Swahili, but only 15 million people worldwide have Swahili as their first language, that means there are many people in Tanzania who speak Swahili as a language other than their first language. This information about Swahili speakers in Tanzania therefore supports the claim that many Swahili speakers in countries where nearly everyone speaks Swahili speak it as a language other than their first language (such as their second, third, or fourth language).

Choice B is incorrect because it doesn't accurately describe information in the table. According to the table, Tanzania has 61 million Swahili speakers, not at most 15 million Swahili speakers. Additionally, the table indicates that 100 percent of Tanzania's population speak Swahili, which means that the number of Swahili speakers in the country and the country's total population should be the same, not that they should differ by such a large amount. Choice C is incorrect because there's no information in the table or the text that indicates where people who speak Swahili as their first language live. Although Kenya's total population can be inferred from the table—if Kenya has 55 million Swahili speakers and 100% of Kenya's population speak Swahili, then Kenya must have a population of 55 million people—nothing suggests that all the people who speak Swahili as their first language live in a single country, let alone that they all live in Kenya. Choice D is incorrect. Although the table does indicate that 100 percent of Kenya's population and 25 percent of the Democratic Republic of the Congo's population speak Swahili, this comparison is irrelevant to the claim that Swahili isn't the first language of many of its speakers even in countries where almost everyone speaks Swahili. On its own, a difference in the proportions of the population who speak Swahili cannot reveal whether those Swahili speakers have Swahili as their first language or a subsequent language.

Q6

D

- D. Most of the folklore that the ethnographers collected consisted of *corridos*—ballads about history and social life—of a clearly recent origin.
-

Choice D is the best answer because it presents a finding that, if true, would support Paredes's argument that Mexican-American folklore is mostly the result of cultural interactions in Mexico and the United States rather than an adaptation of Spanish folklore. The text describes a disagreement among scholars about whether Mexican-American folklore mostly derived from the folklore of Spain or originated in Mexico and the United States as cultures there have interacted. The latter view is the argument that Paredes puts forward. If Mexican-American folklore collected in the twentieth century mostly consists of ballads about history and social life that originated recently, then that would support Paredes's argument, since it would suggest that the folklore mostly arose after Spanish rule ended in the early nineteenth century and that the folklore reflects cultural interactions in Mexico and the United States rather than traditions from Spain.

Choice A is incorrect because the inclusion of songs influenced by sixteenth-century Spanish poetry among Mexican-American folklore collected in the twentieth century would not support Paredes's view that the folklore was the result of cultural interactions in Mexico and the United States rather than an offshoot of Spanish folklore. If anything, the presence of such songs among the folklore collected in the twentieth century would weaken Paredes's argument, since it would reflect the influence of Spanish culture on the folklore. Choice B is incorrect because the mere presence of similarities in Mexican-American folklore across regions would not be sufficient to draw a conclusion about where the folklore originated, let alone to support Paredes's argument that the folklore reflects cultural interactions in Mexico and the United States. In fact, since Paredes argued that Mexican-American folklore is the product of various cultures interacting in Mexico and the United States, he would likely expect there to be regional variations in folklore as different cultures have interacted in different places. Choice C is incorrect because scholars' previous ignorance of the folklore would have no bearing on Paredes's argument that Mexican-American folklore mostly reflects cultural interactions in Mexico and the United States. The folklore's origins are independent of scholars' knowledge of the folklore.

Q7

B

- B. The lowest performing ETL applied through spray coating had a higher power conversion efficiency than the highest performing ETL applied through spin coating.
-

Choice B is the best answer because it describes data from the graph that support Taylor and colleagues' conclusion that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells. The text explains that perovskite solar cells' efficiency at converting light into electricity is diminished by their electron transport layer (ETL), which is applied through spin coating, but that Taylor's team devised a new spray coating method for applying the ETL that improves its power conversion efficiency. The graph displays data on the power conversion efficiency of solar cells in tests conducted by Taylor's team, with bars for both the highest- and lowest-performing ETLs in two data categories: spray coating and spin coating. According to the graph, the lowest-performing ETL applied through spray coating had a power conversion efficiency of between 14% and 16%, while the highest-performing ETL applied through spin coating had a power conversion efficiency of less than 14%. These data confirm that ETLs applied through novel spray coating are more efficient than those applied through traditional spin coating. Thus, the data support Taylor and colleagues' conclusion about spray coating's potential value.

Choice A is incorrect. Although this claim correctly describes the data in the graph by stating that both the lowest-performing ETL applied through spin coating and the lowest-performing ETL applied through spray coating had a power conversion efficiency greater than 10%, this relationship in the data doesn't support or relate to Taylor and colleagues' conclusion that spray coating promises greater efficiency for solar cells than traditional spin coating does. Choice C is incorrect. This claim does address the greater power conversion efficiency of the highest-performing ETL applied through spray coating, compared with the highest-performing ETL applied through spin coating. However, it also incorrectly cites the value for the efficiency of the highest-performing ETL applied through spray coating as approximately 13%, instead of a value between 14% and 16%, and the value for the efficiency of the highest-performing ETL applied through spin coating as approximately 11%, instead of a value between 12% and 14%, as shown in the graph. Choice D is incorrect because Taylor and colleagues' conclusion is based on the difference in the power conversion efficiency of ETLs applied through spray coating and that of ETLs applied through spin coating, not on the difference between the highest- and lowest-performing ETLs applied through just spray coating.

Q8

D

- D. ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice D is the best answer because it most effectively uses data from the graph to complete the student's conclusion about banana ripening time with and without ethylene at different temperatures. The graph shows that at 20°C, the gap between the two bars showing ripening times for ethylene-treated bananas and untreated bananas crosses fewer than 2 gridlines (from about 4 days for ethylene-treated bananas to about 5.5 days for untreated bananas). Meanwhile, the graph shows that at 14°C, 16°C, and 18°C, the gap between the bars crosses more than 2 gridlines (from about 8 days for treated bananas to about 11 days for untreated bananas at 14°C; from about 6 days for treated bananas to about 9.5 days for untreated bananas at 16°C; and from about 5.5 days for treated bananas to about 8.5 days for untreated bananas at 18°C). Since the gap between the bars at each of these temperatures crosses more than 2 gridlines, and since each of these gaps is larger than the gap between the bars at 20°C, it can be concluded that ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice A is incorrect because the graph shows that ethylene-treated bananas stored at 20°C ripen more quickly than ethylene-treated bananas stored at the other temperatures do (about 4 days at 20°C vs. about 5.5, 6, and 8 days at 18°C, 16°C, and 14°C, respectively) and that untreated bananas stored at 20°C ripen more quickly than untreated bananas stored at the other temperatures do (about 5.5 days at 20°C vs. about 8.5, 9.5, and 11 days at 18°C, 16°C, and 14°C, respectively). The information in the graph therefore indicates that storing bananas at 20°C speeds up ripening time relative to storing bananas at the other temperatures shown, not that this storage temperature slows ripening time. Choice B is incorrect because the graph shows that as temperature increases, the ripening time of untreated bananas decreases, from about 11 days at 14°C to about 5.5 days at 20°C, with no exceptions to this trend. The graph therefore shows that differences in temperature were associated with absolute differences in ripening time, not that there was no association between differences in temperature and differences in ripening time. Choice C is incorrect because the graph shows that ripening times of ethylene-treated bananas at 14°C and 16°C were about 8 and 6 days, respectively, which is greater than, not less than, ripening times of ethylene-treated bananas at 18°C and 20°C, which were about 5.5 and 4 days, respectively. In other words, bananas treated with ethylene ripen more slowly, not faster, at 14°C and 16°C than at 18°C and 20°C.

Q9

C

- C. “In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was swiftly completed by the summer.”
-

Choice C is the best answer. The fact that Henry VIII commissioned a survey of church holdings just before enacting the Dissolution of the Monasteries suggests that he was interested in the potential profits from claiming their assets. This supports the historians’ assertion.

Choice A is incorrect. This choice describes a potential effect of the Dissolution of the Monasteries. The decrease in the proportion of monks in England isn’t relevant to the question of Henry VIII’s motivations. Choice B is incorrect. Details about how the monks were treated during the dissolution don’t address Henry VIII’s motivations for enacting the policy in the first place. That the monks could keep the content of their cells (their rooms) and sell off the timber they’d harvested has no impact on the value of the monasteries’ estates—their land holdings. Choice D is incorrect. This choice mentions one impact that the Dissolution of the Monasteries contributed to, two years after it happened. But it doesn’t help explain why Henry VIII might have wanted to enact the policy in the first place.

Q10

D

- D. Although Plato's writings typically characterize representational art as an inferior reflection of the physical world, Aristotle suggests that mimesis can refer to art's capacity to envision hypothetical conditions that could, but don't yet, exist.

Choice D is the best answer because it presents a statement that would support Halliwell's claim that "imitation" is an overly simplified translation of the word "mimesis" in the context of ancient Greek philosophical discussions of representational art and that, because it suggests a view of art as merely copying things that exist in reality, the translation obscures the fact that Plato, Aristotle, and other ancient Greeks understood the relationship between art and reality in varied ways. If Plato's works tend to treat representational art as an inferior or inadequate reflection of the physical world (consistent with the idea of mere imitation), while Aristotle's works suggest that the term "mimesis" can refer to art's ability to show conditions that don't currently exist but could exist (going beyond imitation into novel creation), that evidence would support the claim that the term "mimesis" means more than just "imitation" and that ancient Greek philosophers held different views of the relationship between art and reality.

Choice A is incorrect because information about the root word *mimos* first being used in drama and the root *mim-* coming to be associated with music and poetry wouldn't indicate anything about the meaning of the specific term "mimesis" as it was used by ancient Greek philosophers in discussions of representational art; thus, the information would have no bearing on Halliwell's claim that a common translation of the specific term is an oversimplification. Choice B is incorrect because the issue of art's psychological effects on audiences gets at how people respond to works of art instead of how art itself is related to reality, so the idea that Plato and Aristotle both addressed such effects in their considerations of mimesis wouldn't have any bearing on Halliwell's claim that a common translation of that term oversimplifies the relationship between art and reality. Further, this idea would touch on one general similarity in approaches to mimesis instead of illustrating the multifaceted, or varied, ways Halliwell claims ancient Greek philosophers understood art's relation to reality. Choice C is incorrect because the idea that Plato discussed aesthetics in his early works but didn't use the term "mimesis" until later doesn't give any indication of how Plato or any other ancient Greek philosophers understood the relation between art and reality, so it would have no bearing on Halliwell's claim that a common translation of the term "mimesis" is oversimplified and fails to reflect the varied ways the philosophers understood that relationship.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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